

VENKATA PAVAN KANDULA

DevOps Engineer II | SRE | Kubernetes & Cloud Infrastructure Specialist

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 [Portfolio](#) |  [GitHub](#) |  [LinkedIn](#)

PROFESSIONAL SUMMARY

Results-driven DevOps Engineer with 4+ years of hands-on experience scaling and securing enterprise Kubernetes infrastructure on AWS EKS, implementing GitOps workflows with ArgoCD, and managing high-throughput datastores (Elasticsearch, RDS, DocumentDB) and messaging systems (RabbitMQ, Kafka). Proven track record at TCS managing 100+ microservices across multi-environment promotion chains, with deep expertise in Terraform IaC, GitHub Actions CI/CD, and observability stacks (Datadog, Grafana, ELK). Delivered a 40% reduction in release cycle time and eliminated 8–10 hours of weekly manual toil through Python/Bash automation.

CORE TECHNICAL SKILLS

- **Kubernetes / EKS:** EKS, GKE, AKS cluster design, RBAC, network policies (Calico/Flannel), Istio service mesh, Helm, pod security standards, ArgoCD GitOps
- **CI/CD:** GitHub Actions (primary), GitLab CI/CD, Jenkins, AWS CodePipeline, ArgoCD — multi-env pipelines Dev → QA → UAT → Prod
- **Cloud (AWS):** EC2, EKS, S3, RDS, Aurora, DocumentDB, Lambda, API Gateway, IAM, VPC, CloudWatch, Secrets Manager
- **Infrastructure as Code:** Terraform (modules, workspaces, remote state), Helm chart authoring, Ansible, AWS CloudFormation, GitOps workflows
- **Datastores:** Elasticsearch (ELK), AWS RDS, Aurora (Multi-AZ, read replicas), DocumentDB, PostgreSQL, Databricks Lakehouse
- **Messaging Systems:** RabbitMQ, Kafka — queue management, throughput tuning, dead-letter queue configuration
- **Monitoring & Observability:** Datadog, Grafana, Prometheus, ELK Stack, CloudWatch, Splunk — real-time dashboards & alerting
- **Scripting & Automation:** Python (self-healing systems, predictive scaling, automation), Bash/Shell, Groovy (Jenkins shared libraries)
- **Networking & Security:** VPC design, ALB/NLB, DNS, TLS/mTLS, OPA/Gatekeeper, zero-trust architecture, container image scanning
- **AI in DevOps:** Python-driven predictive scaling, self-healing Kubernetes operators, automation-first approach

PROFESSIONAL EXPERIENCE

Senior DevOps / Site Reliability Engineer

Tata Consultancy Services (TCS) — Hyderabad, India | Mar 2022 – Present (4 Years)

- **Kubernetes Cluster Management (EKS/GKE/AKS):** Designed, provisioned, and managed production-grade Kubernetes clusters on AWS EKS supporting 100+ microservices; enforced RBAC, network policies, CNI plugins (Calico/Flannel), Istio service mesh integration, and pod security standards; performed cluster upgrades with zero downtime.
- **ArgoCD GitOps Deployments:** Implemented and maintained ArgoCD-based GitOps workflows for consistent, automated application deployments across Dev → QA → UAT → Prod environments; ensured single source of truth for all Kubernetes manifests and Helm charts, reducing config drift and rollback time.
- **Datastore Management (Elasticsearch, RDS, DocumentDB):** Administered Elasticsearch clusters for the ELK observability stack; managed AWS RDS (PostgreSQL, Aurora Multi-AZ with read replicas) and DocumentDB; performed performance tuning, backup policies, and scaling operations to support high-throughput workloads.
- **Messaging Systems (RabbitMQ, Kafka):** Configured and managed RabbitMQ and Kafka queuing systems for event-driven microservices architectures; tuned throughput settings, monitored queue depths, implemented dead-letter queue strategies, and ensured high availability for mission-critical pipelines.

- **Terraform Infrastructure as Code:** Authored modular Terraform configurations for all AWS infrastructure (VPC, EKS, RDS, S3, IAM, Lambda, API Gateway); used workspaces and remote state for environment isolation; maintained GitOps-aligned IaC repositories enabling repeatable, auditable provisioning.
- **GitHub Actions CI/CD Pipelines:** Designed and maintained end-to-end CI/CD pipelines using GitHub Actions, GitLab CI/CD, and Jenkins; integrated automated build, test, security scanning, and deployment stages — achieving a 40% reduction in release cycle time; on-boarded multiple applications onto the DevOps toolchain.
- **Monitoring & Observability (Datadog / Grafana / ELK):** Deployed enterprise observability stack using Datadog, Prometheus + Grafana, and ELK Stack; built real-time dashboards for cluster health, pipeline metrics, and datastore performance; implemented intelligent alerting that significantly reduced MTTR.
- **Scalability & Architecture Solutions:** Partnered with engineering teams to diagnose and resolve infrastructure scalability bottlenecks; designed auto-scaling strategies using HPA/VPA for Kubernetes workloads and AWS Auto Scaling for EC2; implemented predictive scaling via Python automation.
- **Python & Bash Automation:** Developed Python and Bash scripts eliminating 8–10 hours of weekly manual toil for infrastructure health checks, artifact management, and operational reporting; built self-healing Kubernetes operators for automatic workload recovery and incident remediation.
- **Engineering Team Support & On-Call:** Provided 24/7 on-call support; resolved P1/P2 incidents within defined SLA; conducted RCA for critical failures; mentored junior engineers on Kubernetes, Terraform, and CI/CD best practices; produced architecture diagrams and operational runbooks.

KEY ACHIEVEMENTS

- 40% reduction in release cycle time through end-to-end CI/CD pipeline optimization across GitHub Actions, GitLab CI/CD, and Jenkins.
- Eliminated 8–10 hours of weekly manual toil via Python/Bash automation for infrastructure health checks, artifact management, and reporting.
- Zero-downtime application on-boarding — successfully on-boarded multiple applications onto ArgoCD GitOps with 100% error-free, zero-production-incident record.
- Enterprise observability implemented using Datadog + Prometheus + Grafana + ELK Stack, driving significant MTTR reduction across production Kubernetes infrastructure.
- Scalability uplift — designed and deployed HPA/VPA-based auto-scaling for 100+ microservices on EKS, ensuring SLA adherence during traffic spikes.
- Recognized at TCS for driving CI/CD transformation and Kubernetes infrastructure automation initiatives organization-wide.

EDUCATION

Bachelor of Technology (B.Tech) — Mechanical Engineering

Aditya Engineering College, JNTU Kakinada | Graduated October 2020